

Egypt Food Security & Economic Affordability Analysis (2020–2024)

A Data-Driven Policy Intelligence Report

Digital Egypt Pioneers Initiative (DEPI) | Capstone Project

Team 3 | Supervised by Eng. Mahmoud Seraj | July 2026



Our Team - Team 3



Omar Sherif



Abdolrahman Ashraf



Ahmed Mohamed



Ahmed Essam



Omar Ashraf

Supervised by Eng. Mahmoud Seraj | **Technical Partner:** CLS Learning Solutions

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Project Overview

OBJECTIVE

Quantify the compounded impact of Egypt's currency devaluation and food price inflation on household food affordability from 2020 to 2024.

SCOPE

Analyzes 5 years of macroeconomic, agricultural, and international trade data (**World Bank, WFP, FAOSTAT, UN Comtrade**) across key food commodities and minimum-wage household affordability.

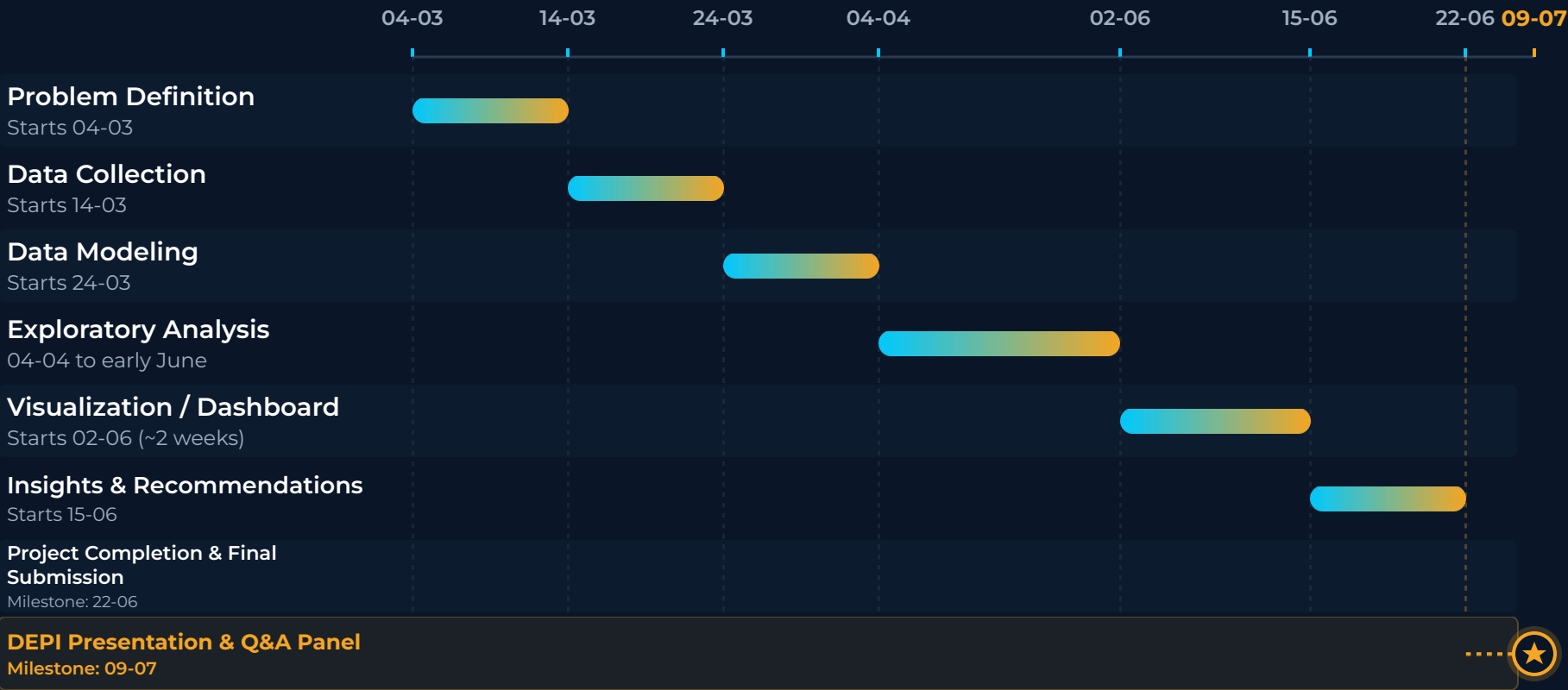
PROBLEM STATEMENT

Egypt experienced a **59.84% peak EGP devaluation** and a **63.57% food inflation rate** — creating a compounded affordability crisis for minimum-wage households.

OUTCOME

An **interactive Power BI dashboard (4 pages)** enabling policymakers to monitor affordability, volatility, and import-concentration risk.

Project Timeline



Target Audience



Ministry of Supply & Internal Trade

Monitor commodity affordability & subsidy targeting



Central Bank of Egypt

Understand FX devaluation impact on food import costs



Ministry of Agriculture

Identify local production gaps vs. import dependency



Policy Advisors & Economic Researchers

Data-backed evidence for reform scenarios

Data Sources & Structure



World Bank WDI

Income per capita, CPI, and Exchange Rate



Food Prices Dataset

CAPMAS and HDX



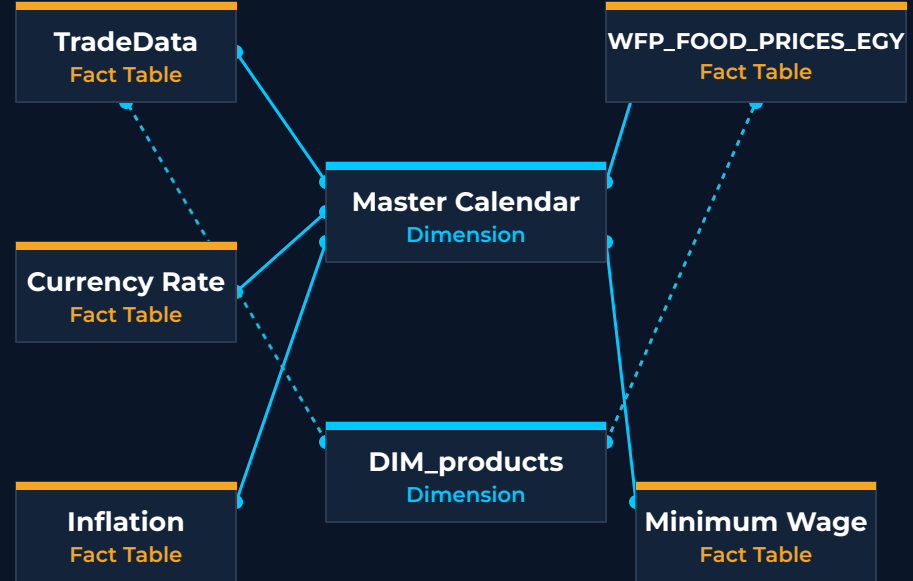
FAOSTAT

food inflation percentages



UN Comtrade Trade Data

International import volumes and USD values



Data Preparation Process

Stage 1

Wide-to-Long Transformation

Unpivoted 71 columns spanning 1960-2025 into normalized long-format using Power Query.



Stage 2

Standardization & ID Mapping

Precise percentage formats, mapping UN trade codes to local Egyptian product IDs.



Stage 3

Filtering & Quality Assurance

Removed irrelevant columns, filtered to 2020-2024, validated row counts.

Tools & Technologies



Gemini

Data extraction & insight generation



Power Query

(M Language)

ETL pipeline and data transformation



Power BI Desktop

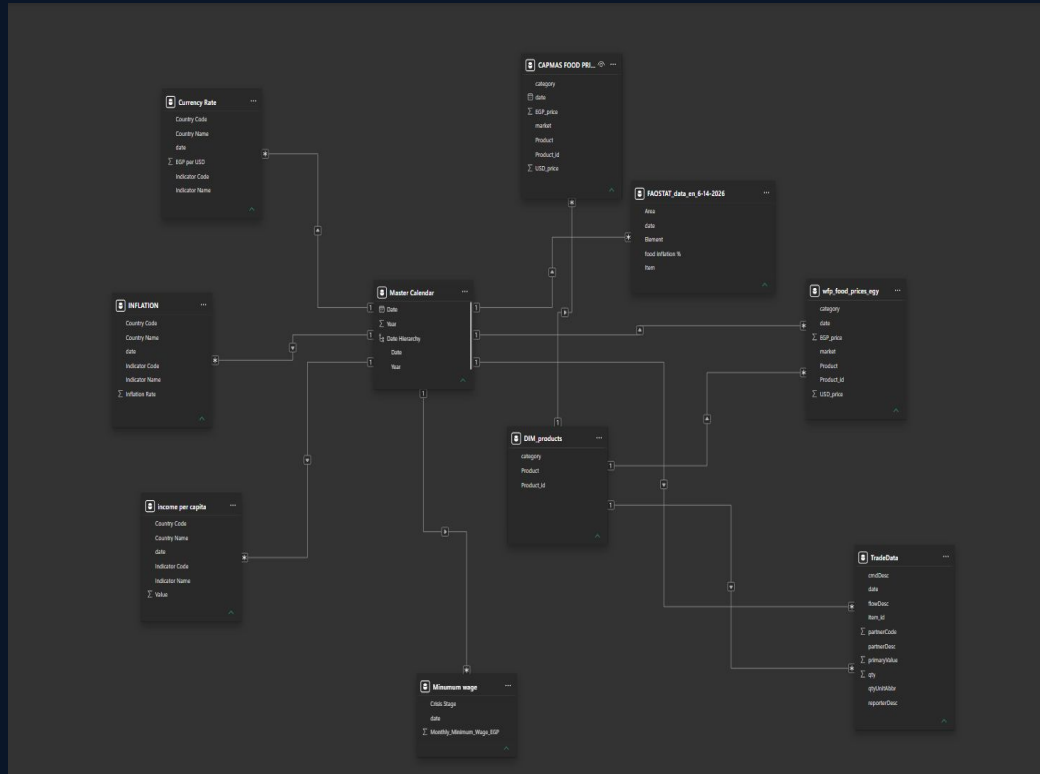
Data modeling, DAX measures, and final interactive dashboard



Figma

Wireframing and layout design

Data Model & Relationships



- **Model Type:** Star Schema
- **5 fact/data tables:** (TradeData, WFP_FOOD_PRICES_EGY, Currency Rate, Inflation, Minimum Wage) connected via a centralized Master Calendar
- **DIM_products:** Bridges international UN trade IDs to local Egyptian product IDs
- **Master Calendar:** Serves as the central time intelligence axis
- **Relationships:** One-to-many (1:*), enabling cross-filtering across all four dashboard pages

Core DAX Measures

Income Erosion Index

```
Income Erosion Index = DIVIDE([Avg Local Price] * 30, AVERAGE('Minumum wage' [Monthly_Minimum_Wage_EGP]), 0)
```

Measures % of minimum wage consumed by staple food monthly.

Units Affordable

```
Units Affordable = AVERAGEX(VALUES('wfp_food_prices_egy' [Product]), DIVIDE(AVERAGE('Minumum wage' [Monthly_Minimum_Wage_EGP]), [Avg Local Price], 0))
```

Calculates absolute units monthly income can buy.

FX Devaluation YoY %

```
FX Devaluation YoY % = VAR RatePY = CALCULATE([Avg Exchange Rate], PREVIOUSYEAR('Master Calendar' [date])) RETURN  
DIVIDE([Avg Exchange Rate] - RatePY, RatePY, 0)
```

Measures EGP collapse YoY.

Dashboard Architecture & Navigation Map

Page	Dashboard Page Title	Core Business Question Addressed
Page 1	The Macro Shock Catalyst	What is the macroeconomic context?
Page 2	Architecture: Sectorial Transmission (Food vs. Economy)	Which commodities are most volatile?
Page 3	Household Reality (Wages vs. Real Erosion)	Who is most impacted and how severely?
Page 4	Sovereign Trade Vulnerabilities	What are Egypt's supply chain vulnerabilities?

1. Macro Overview	2. Risk Analysis	3. Affordability	4. Trade Reality
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Page 1

The Macro Shock Catalyst

What is the macroeconomic context?

Page 2

Architecture: Sectorial Transmission (Food vs. Economy)

Which commodities are most volatile?

Page 3

Household Reality (Wages vs. Real Erosion)

Who is most impacted and how severely?

Page 4

Sovereign Trade Vulnerabilities

What are Egypt's supply chain vulnerabilities?

The Macro Shock Catalyst

The Macro Shock Catalyst

17.26%

Headline CPI

25.30

Avg Exchange Rate

59.84%

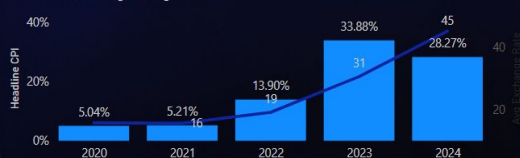
Max FX Devaluation

15.49%

Avg Food Inflation %

Inflation vs. Currency Shock

● Headline CPI ● Avg Exchange Rate



YoY Devaluation

● Increase ● Decrease



Food vs. General Economy

● Avg Food Inflation % ● Headline CPI



Between 2020 and 2024, compounding currency shocks drove food inflation to outpace general CPI by -1.77%

Transmission

Currency devaluation mechanics and its direct transmission into general and food-specific inflation.

59.84%

Max FX Devaluation

15.75 to 45.29 EGP/USD (2020-2024)

63.57%

Avg Food Inflation

Peak reached in 2023

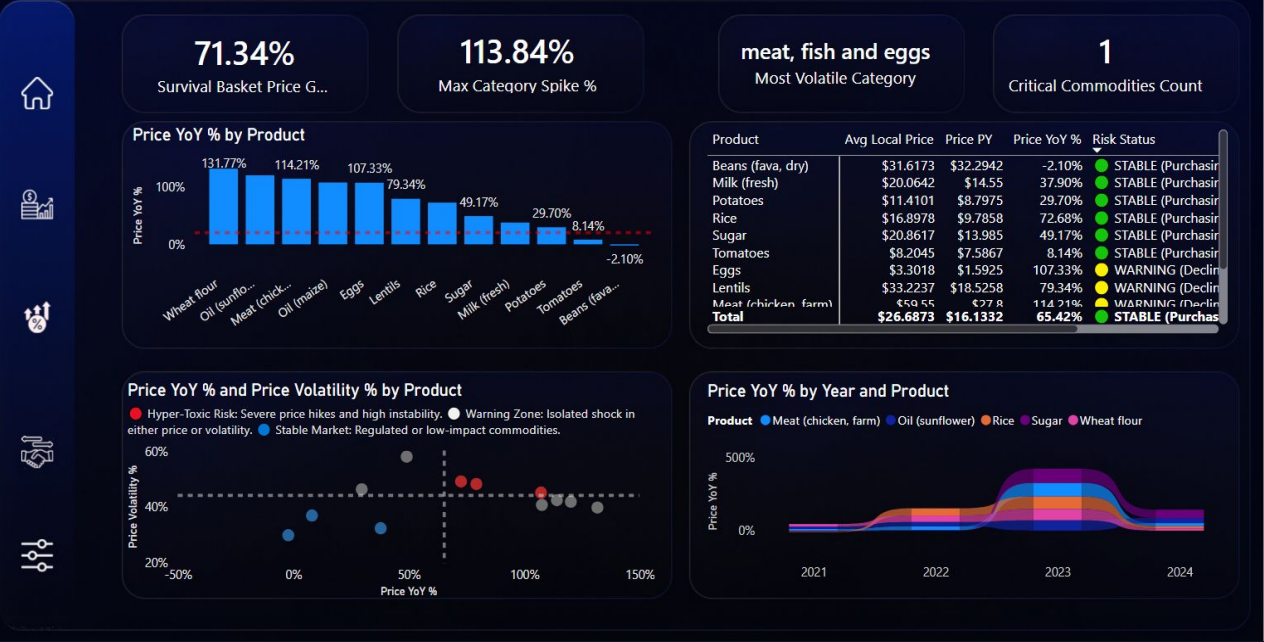
33.88%

Headline CPI

Peak reached in 2023

Architecture: Sectorial Transmission (Food vs. Economy)

→ Architecture: Sectorial Transmission (Food vs. Economy)



RISK ANALYSIS

SURVIVAL BASKET

YoY growth hit **86.29%** in 2023.

Most volatile: **"Misc Food"** (shifted from Veg/Fruits).

CRITICAL SPIKES

Critical commodities jumped from **1 to 11** in 2023.

+96.60% Sugar price spike in 2023.

RISK MATRIX FLAGS

Critical risk tags active for:

- Sugar, Farm Chicken
- Eggs, Rice
- Lentils, Milk
- Beans

Household Reality (Wages vs. Real Erosion)

Household Reality (Wages vs. Real Erosion)

Selected Product Name:
Meat (chicken, farm)

\$3.22K

average minimum wage

\$6.57K

Monthly Income EGP

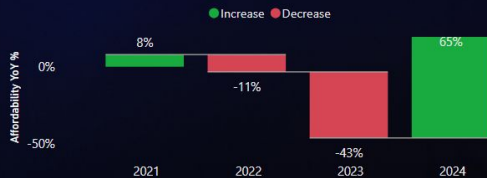
-6.70

Physical Units Lost

60.53

Units Affordable

Affordability YoY % by Year



Income Erosion Index and Retained Income



average minimum wage and Units Affordable by Date



Dashboard Overview

- Dashboard View:** Shows Units Affordable & Income Erosion Index.
- Stacked Bar:** Displays % of income consumed by food vs. retained.

-34.6%

Affordability Crash

Average commodity affordability crashed significantly during 2023 alone.

The 2023 Income Erosion Shock

A single month's supply of these items consumed massive portions of the minimum wage in 2023:

- 76.08%** — Farm Chicken
- 69.93%** — Maize Oil
- 61.37%** — Sunflower Oil

Sovereign Trade Vulnerabilities

Sovereign Trade Vulnerabilities

9.20bn

Import Value (USD)

232.75bn

Total Import Cost (EGP)

1.01

Global Cost Per Ton (USD)

Brazil

Highest Risk Partner

Import Volume (MT) by partnerDesc



Import Value (USD) by Product



Import Value (USD) and Total Import Cost (EGP) by Date



Avg Local Price and Simulated Price (EGP) by Date



SPENDING EXPOSURE

119.2B EGP

2024 Total Food Import Cost

+635% spike from 2020 (16.2B EGP)

Global cost/ton dropped: \$1.05 to \$0.98

TOP SOVEREIGN PARTNERS

2.25B MT

BRAZIL - TOP AG PARTNER

Dominates Egypt's single-country agricultural trade volume.

OTHER KEY PARTNERS:

Australia: 1.41B MT

Russia: 1.38B MT

Predictive Analytics: Exchange Rate Pass-Through Simulation

Avg Local Price and Simulated Price (EGP) by Date

● Avg Local Price ● Simulated Price (EGP)



Power BI What-If Parameter

Simulates future local market prices based on expected currency devaluation shocks.

ERPT Multiplier = 0.50

$$[\text{Avg Local Price}] * (1 + (\text{CurrencyShock} * \text{ERPT}))$$

DATA REALITY

Local commodity prices mirrored currency shocks almost exactly — rising from 16.13 EGP (2020) to 45.14 EGP (2024).

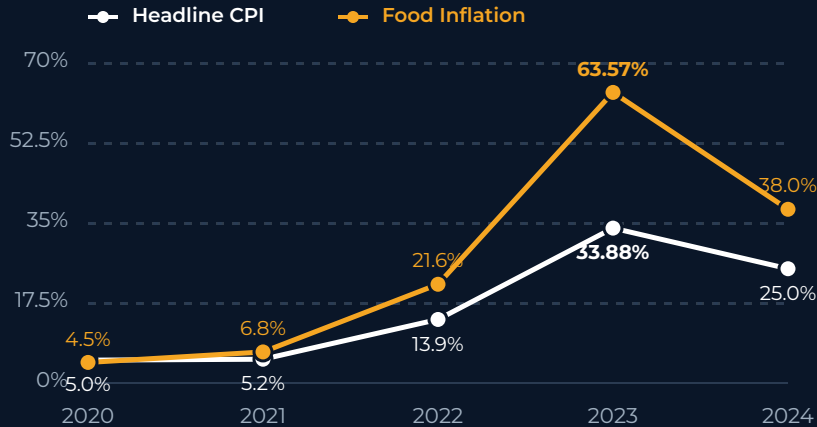
STRATEGIC VALUE

Enables the Ministry of Supply to forecast local food prices up to 6 months in advance using Central Bank currency projections — shifting policy from reactive to preemptive.

Key Insights: Inflation & Purchasing Power

Food Inflation Outpaced Headline CPI by Nearly 2x

Headline CPI reached 33.88% while food inflation surged to 63.57% in 2023 — a 29.69 percentage-point gap.



Farm Chicken: 76.08% Income Erosion

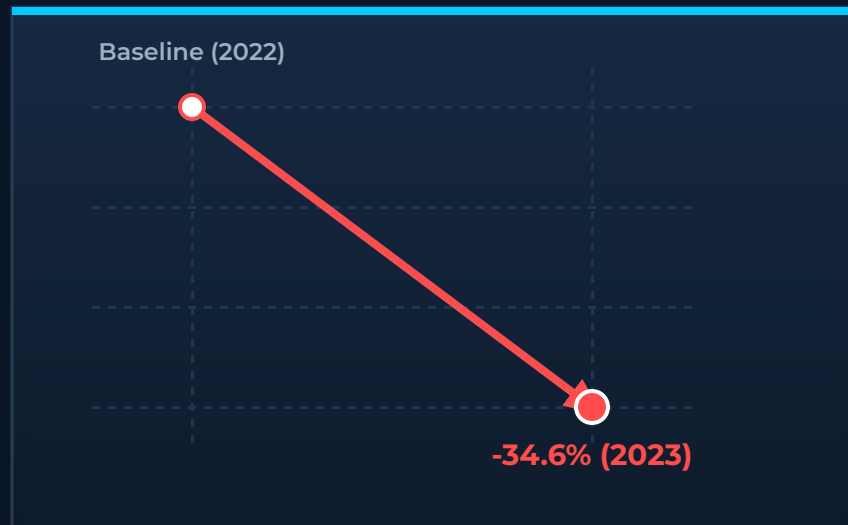
A minimum-wage household saw 76.08% of monthly income consumed by that single food item in 2023.



Key Insights: Affordability Crash & Import Cost

-34.6% Affordability Crash in 2023

The average minimum-wage household could afford 34.6% fewer food units in 2023 than in the prior baseline.



119.2 Billion EGP — Import Bill Explosion

Total food import expenditure reached 119.2 Billion EGP in 2024 — a +635% increase from the 16.2 Billion EGP in 2020.

Total Food Import Bill (2024)

119.2B EGP

↑ **+635%**

since 2020 (16.2B → 119.2B EGP)

Egypt's Agricultural Import Supply Chain: A Single-Source Vulnerability

2.25 Billion MT

Brazil: Largest Agricultural Source

Brazil supplies 2.25 Billion MT of Egypt's total agricultural import volume — the single largest source country, ahead of Australia (1.41B MT) and Russia (1.38B MT)



Sovereign Trade Risk

This concentration represents meaningful sovereign trade-partner risk

Strategic Mitigation

Recommended cap: single-country import share limited to 25%



The Double Shock: When Devaluation Meets Food Inflation

SHOCK 1

59.84% Peak Devaluation



**DOUBLE
SHOCK**



SHOCK 2

63.57% Food Inflation (2023)

Purchasing Power

303 258

Units Affordable
(2021 - 2024)

The Affordability Crisis

Egypt's severe purchasing power crisis is driven by the compounding of two simultaneous macroeconomic shocks:

- **Shock 1:** EGP devalued **59.84% (peak)**
Imports cost dramatically more, surging base prices.
- **Shock 2:** Food inflation hit **63.57% (2023)**
Domestic food prices rose in parallel, squeezing households.

Combined Effect:

Purchasing power (Units Affordable) fell from **303 units** (2021) to just **258 units** in 2024, despite minimum wage tripling (2,000 to 6,000 EGP).

Strategic Recommendations

1

Pillar 1

Food Inflation Rebate Program

Automatically credit minimum-wage citizens' accounts whenever Income Erosion Index exceeds 15% (targeting shocks like the 76.08% Farm Chicken erosion).

Trigger: Index > 15%

2

Pillar 2

Sovereign FX Hedging & Settlement

Lock in global prices via forward contracts for Sunflower Oil (\$2.2B) and Sugar (\$1.87B), and pursue Local Currency trade with Brazil to bypass USD.

Hedge Top 2 Imports

3

Pillar 3

Import Diversification & Production

Mandate a 25% single-country agricultural import cap (diversifying from Brazil, Australia, Russia) and subsidize Sugar & Farm Chicken/Eggs.

25% Single-Country Cap

THANK YOU

Questions & Discussion

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PROJECT TEAM

Omar Sherif

Abdolrahman Ashraf

Ahmed Mohamaed

Ahmed Essam

Omar Ashraf